01. Ligand Benchmark Report

Experiment ID: P4.2-LIGAND-FOUNDATION **Model Type**: foundation_ligand **Molecular Model**: CHEMBERTA

Executive Summary

This report summarizes the candidate molecular foundation sweeps. The candidate model (**ChemBERTa** hybrid fusion GNN) achieved a validation mean RMSE of **1.5034**, outperforming the **Phase 4.1 Protein Foundation** baseline (1.5269 RMSE).

Performance Matrix

Seed	Phase 4.1 Protein Foundation (Baseline)	GNN + ChemBERTa (Candidate)	Difference
42	1.5124	1.4872	-0.0252
123	1.5342	1.5098	-0.0244
777	1.4421	1.4210	-0.0211
2024	1.6212	1.5982	-0.0230
3407	1.5245	1.5008	-0.0237
Mean	**1.5269**	**1.5034**	**-0.0235**

Benchmark Studies

Study A: Foundation Model Comparison

Compare ChemBERTa vs. MolFormer (pooling `Graph Node Mapping`, 100% data):

- **ChemBERTa**: Mean RMSE = 1.5034
- **MolFormer**: Mean RMSE = 1.5116

Study B: GAT Pooling strategy

- **Graph Node Mapping** (Default): Mean RMSE = 1.5034 (Maintains atom-level spatial topology).
- **Atom Mean**: Mean RMSE = 1.5486
- **CLS**: Mean RMSE = 1.5849

Study C: Sample Efficiency Sweep

Evaluate performance across training fractions (10%, 25%, 50%, 100%):

- **Phase 4.1 Baseline**: 10% data = 1.691 | 100% data = 1.527
- **Phase 4.1 + ChemBERTa**: 10% data = 1.614 | 100% data = 1.5034
- **Phase 4.1 + MolFormer**: 10% data = 1.625 | 100% data = 1.5116